

Category	Score (0–5)	Criteria
Introduction & Related Work	4	The introduction and related works are well written. However, the paper did not addresss whether linearization of the Ising model has been approached before.
Professionalism	5	Few grammatical errors. Could benefit from subsection headings within the introduction.
Total	9 /10	

Table 1: Peer-Review Evaluation Summary

1 Summary

The paper investigates three notions of linearization of a stochastic, high dimensional, non-linear system, specifically that of the Ising model. The first notion is a standard linearization of the Ising model, which

The paper as is suggests that future sections will then highlight the differences in equilibria and bifurcation behavior between the three notions of linearization.

2 Criticisms

2.1 Introduction & Related Work

The paper did not address whether linearization of the Ising model has been approached before. If no work on linearization of the Ising model has been done, the paper should state that. If work has been done, the paper should reference the work and discuss the differences in the approaches.

The end of the introduction is a little unclear. The second to last paragraph implies that "renormalization" will be an approach used to linearize the Ising model. However, the last paragraph implies that linearization is on the other end of the spectrum from "renormalization". This is confusing, particularly because the Methodology section refers to the Fokker–Planck / Projection Approximation as one of the linearization concepts/tools. Within the Fokker–Planck / Projection Approximation section, the method is described as a coarse-graining method, which in the Introduction was described as a renormalization approach. Thus, the Fokker–Planck / Projection Approximation is presented as a renormalization approach, which, as stated earlier, is presented as not a linearization approach.

2.2 Professionalism

In the first paragraph of the introduction, it would be helpful to introduce the idea of an alignment preference in its own sentence before this one: "Atoms have an associated energy level, which is higher for the spin that conflicts with their alignment preferences (depending on the alignment of their neighbors and any external forces)"

Typo in fourth paragraph (correction contained within brackets): "Although the Ising model was first proposed to model the spins of atoms in a material, the properties of the model, including phase transitions and a critical [point (?)] [3], it and essentially isomorphic models are used to model percolation processes phase transitions [5], and forest fires [6], to name only a few"

Additionally, the phrase "the properties of the model, including phase transitions and a critical [point (?)] [3]," referenced previously should be removed from the sentence or the rest of the sentence be modified for grammatical correctness.

Due to the length of the introduction, the section could benefit from subsection headings. For example, "Motivation for the use of the Ising model" could be one subsection.